Measuring yield shear strains of fresh 3D printable mortars with DIC Pocket Shear Vane tests: sample size and vane influence zone study

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Abstract. Quality control is critical for extrusion-based 3D-printed concrete, where in-situ rheological characterization remains challenging. Among other tests, the *Pocket Shear Vane* (PSV) was recently proposed to measure the early age evolution of material strength (or yield stress), referred to as structuration. However, it lacks a standardized protocol, particularly regarding sample size. In this paper, we introduce a *Motorized and digital PSV integrating Digital Image Correlation* (MPSV-DIC), a novel apparatus that automates and precisely controls PSV test conditions, while monitoring strains at the sample's top surface with an embedded camera. First, the method is validated by comparing results on printable mortars against rheometer tests, verifying both the measurements and that Couette-like yield conditions are satisfied if the specimen is sufficiently large. While the strain localization at the vane edge is confirmed, the MPSV-DIC observations further illustrate the shear strain diffusion around the vane, highlighting an influence zone. These observations provide first sample size recommendations for PSV testing of printable mortars.

Keywords: extrusion-based concrete 3D printing, pocket shear vane, structuration rate, digital image correlation, sample size

1 Introduction

Since the first proofs-of-concept and technological prototypes, extrusion-based 3D concrete printing (E-3DCP) applications have emerged in recent years, showing potential to reduce labor, costs, and environmental impact. E-3DCP is a layer-by-layer deposition process, introducing successive manufacturing constraints: pumpability, extrudability, and buildability. These relate to the printed material rheological and mechanical properties, which may vary due to variations in its constituents, and also because of environmental and processing conditions. Controlling the fresh material properties is therefore critical to ensure reliable printing, particularly in the context of developing

industrial applications. To do so, several mechanical tests have been proposed. The material strength evolution (called structuration), which relates to printability, may be tracked after deposition by carrying out series of destructive tests on printed material [1], with a general methodology described by the authors elsewhere [2].

Mechanized laboratory tests such as unconfined compression, tension, and direct shear may be applied, however requiring important experimental effort. In-situ manual tests such as fast penetration [3], and various shear tests [2,4,5] such as those carried out with the *Pocket Shear Vane* (PSV) show high potential thanks to their ruggedness and simplicity, which make them usable in a factory environment. While the PSV originates from in-situ geotechnical tests, with a virtually infinite sample size, PSV testing in the E-3DCP factory is carried out on finite-size printed specimens or buckets of printed material, whose sizes are preferably kept to a minimum. Although sample depth has been studied [2], and protocols spacing samples by a distance of approximately the vane diameter have been reported [2,5], no study has examined the effect of sample size in terms of radial distance (or cup size), which would help develop standardized protocols.

In this paper, we introduce a *Motorized and digital Pocket Shear Vane integrating Digital Image Correlation* named MPSV-DIC, a novel apparatus that automates and precisely controls PSV test conditions, while monitoring the sample's top surface strains with an embedded camera. The sample size influence is first investigated by performing shear stress measurements on a printable mortar, which are validated against rheometer tests, for various sample sizes. Then, a method is proposed to identify the geometrical extent of the influence zone where most significant shear strains are distributed in the MPSV-DIC test sample, at the onset of yielding, for various materials. This approach provides first possible explanations of sample size influence and recommendations for PSV testing of printable mortars.

2 Materials and methods

2.1 MPSV-DIC tests

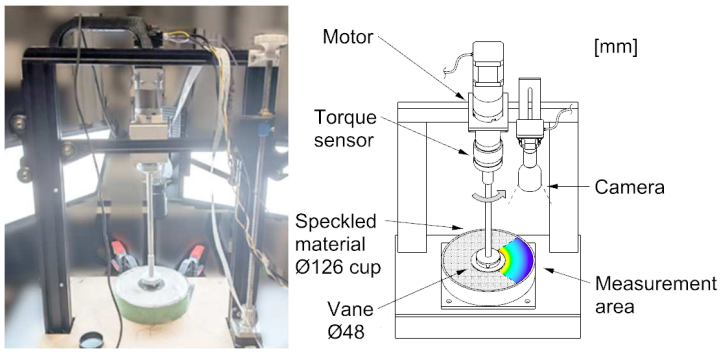


Fig. 1. Photo and schema of the MPSV-DIC device.

As shown in Figure 1, the MPSV-DIC uses the standardized PSV “Sensitive” vane geometry (48 mm diameter, 5.2mm length). The rotational shear testing principle remains identical to that of the PSV; however, actuation and measurements are automated. The device consists of an aluminum frame holding together the sample with vane head pre-inserted, the torque measuring and actuation system, and an optical imaging and recording system for which the data is later postprocessed separately for DIC. The measurement and actuation systems are operated synchronously using an embedded Raspberry Pi computer.

As with the PSV, the vane is rotated while the sample holder remains fixed. However, the PSV soft spring and dial connected to the vane, providing manual force-controlled operation and force measurement, is replaced by a rigid *Interface MRT* torque sensor (with a ±2 Nm measuring range) placed between the vane shaft and a displacement-controlled stepper drive motor. A 1:100 gear reducer is attached to the motor in order to improve torque capacity and angular displacement resolution. The tests are run at a fixed rotational speed of 0.01 RPM.

The imaging system is mono-correlation type, using a color 12.3 MP *Raspberry Pi HD* Camera (*Sony IMX477* Sensor) with a 4-12mm variable focus lens. It is attached to the machine frame with an articulated arm clamp for positioning and a translation stage to adjust the focus. The sample is illuminated by two flood LED lights positioned above, ~45° cross-angle. The raw images are postprocessed for DIC using *GOM Correlate 2019* software.

2.2 Rheometer tests

Rheometer tests are carried out with an *Anton Paar MCR102e* equipped with a 4-blade vane *ST14-4V-35*, 14.04 mm diameter and 35 mm length. The vane is inserted flush with the sample surface, and tests are run at a fixed rotational speed of 0.01 RPM, without pre-shearing.

2.3 Sample preparation and mix design

Sample preparation.

For MPSV-DIC, material is poured into a cylindrical plastic cup of 126 mm diameter and 27 mm height. In order to fit the plane strain hypothesis, the sample surface, especially near the vane, should be as flat as possible: that is why the sample is gently tapped manually to provide a flat surface and remove air bubbles. For the same reason, special attention is given to the vane insertion step, already studied elsewhere [2], where the material might recede near the vane blades due to the insertion motion. As the material is relatively flowable after this tapping step, vanes are inserted manually at the center of the cup by penetrating the material while applying a light rotational oscillatory motion, so that the material would flow around the blades without receding. Then, a black speckle pattern is applied using black matte spray paint. The sample is left at rest for at least 1 min before testing to allow reflocculation.

For rheometer tests, material is poured into cylindrical large plastic cups (126 mm diameter, 44 mm height), or *Anton Paar C-CC27/D/AL* disposable small cups (29 mm diameter, 75 mm height). Vane insertion is done by the rheometer before testing, while the material is gently tapped to fill the recess around the vane blades caused by insertion, and sample left at rest 1 min before testing.

Mix design and material preparation.

Table 1. Mix designs.

Component	Weight (g)	Indicative composition (kg/m^3)
Cement paste, W/C = 0.3		
Water	600.0	486
White Portland Cement	2000.0	1620
Printable mortar UT		
Water	573.2	290
Plasticizer Sika ViscoCrete-1100 NT	13.0	6.6
White Portland Cement	2004.6	1014
Silica sand Marutou Mizunami N°7	2004.6	1014

A cement paste (W/C = 0.3) and an accelerator-free (so-called one-component or 1K) printable mortar were used. The respective compositions are detailed in Table 1, materials were weighed as reported in the weight column for each batch, and an indicative relative composition with constituents in kg/m^3, based on typical density values of constituents, is also reported for comparison with other materials. In order to improve the visual contrast between the base material and the black speckle pattern applied for DIC, it is preferable for the materials to be white. That is why all mixes are based on white cement. The materials were prepared indoors at room temperature as follows:

- For the cement paste, it is dry-mixed for 1 min at low speed in a *Maruto SS-C-413* 5 L planetary mixer to visually check the powder dryness and integrity. Then, the mixer is stopped and water added. The paste is then mixed for 2 min at low speed, then the mixer is stopped and the bowl scraped for approximately 30 seconds to mix possible dry spots. It is finally mixed for 2.5 min at high gear speed.
- For the UT printable mortar, all dry components are weighed directly in the *Marui MIC-362* 20 L planetary mixer bowl, and dry mixed for 1 min at 100 RPM. Then, the mixer is stopped after which the pre-mixed water and plasticizer

mixture is poured in. The mortar is then mixed at 100 RPM for 2.5 min, scraped 30 seconds, mixed at 200 RPM for 2.5 min and 280 RPM for 2.5 min.

2.4 Vane yield shear stress description by the Couette analogy and structuration rate

A common framework to interpret the yield stress τ_{cr} obtained by vane tests such as vane rheometer and PSV is based on the Couette analogy. The material is assumed to yield over an equivalent cylindrical boundary of the vane, with a vertical surface of diameter D and height H, and a bottom disc. The vane is flush with the sample surface, so there is no top disc contribution. The resulting torque T_{cr} is [2]:

$$T_{cr} = \left(\frac{\pi D^3 \tau_{cr}}{12}\right)\left(1 + \frac{6H}{D}\right) \quad (1)$$

The yield stress can be detected as a plateau or peak value in the torque history of the rheometer or MPSV-DIC. Structuration of cementitious materials at a very early age may be described using a linear model where t is the time after rest (or material age), τ_0 is the initial yield stress and A_{thix} the structuration rate [6]:

$$\tau_{cr(t)} = \tau_0 + t \cdot A_{thix} \quad (2)$$

2.5 Plane rotational shear strain measurement with DIC

The determination of the yield shear strain γ of printable mortars is not as straightforward as the yield stress, because of their viscoplastic behavior. The MPSV-DIC experiment provides the sample top surface displacement field and yield stress, which can be postprocessed to compute the Von Mises equivalent strain ε_{vm} field. As the vane test provides a simple shear state, the linearized shear strain γ can be derived from ε_{vm} as [7]:

$$\gamma = \varepsilon_{vm} * \sqrt{3} \quad (3)$$

The difference between this metric and the effective strain is less than 5% for $\gamma < 1$ [8]. In the case of higher strains, metrics accounting for finite deformation may be used.

3 Results and discussion

3.1 Effect of the test method and cup size on the failure mode and yield stress

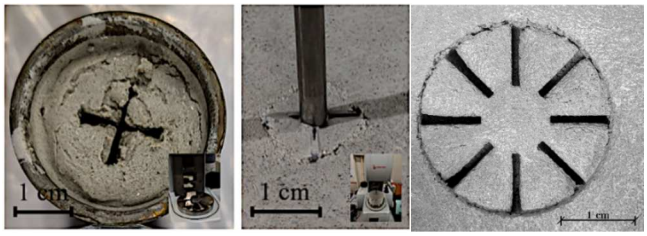


Fig. 2. Appearance of tested printable mortar samples after yielding using the rheometer with small cups (left), large cups (middle) and using a standard PSV (right).

Distinct yielding patterns were observed depending on the testing method. Typical results are illustrated in Figure 2. The large cup pocket shear vane configuration produces a localized cylindrical failure near the vane, that fits well with the theoretical framework of the Couette analogy. Large cup rheometer samples also display localized failure near the vane, but square-shaped. This is likely attributable to the four-bladed design, promoting stress concentration at the blade edges. In contrast, small cup rheometer samples do not show clear Couette-like vane boundary localization, often exhibiting complex crack features mainly located at the cup edges.

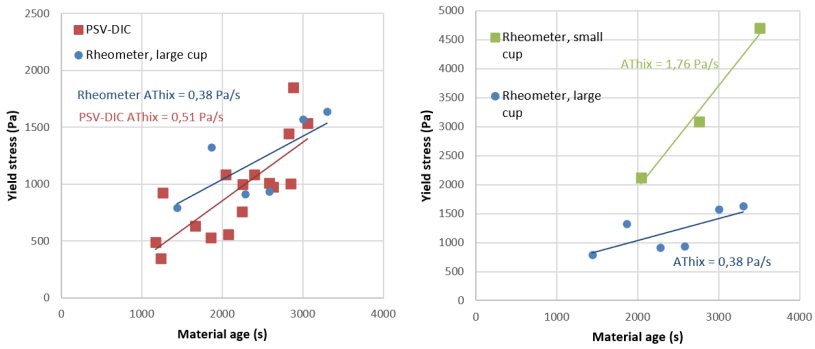


Fig. 3. Yield stress measurements and structuration rate of the printable mortar measured using the rheometer and MPSV-DIC device. Left: Comparison between the large cup rheometer and MPSV-DIC. Right: Comparison with the rheometer, using a small or large cup.

Yield stress and structuration rate measurement results obtained using the three methods are illustrated in Figure 3. Using large cups, the structuration rate obtained from pocket vane and rheometer shows a consistent trend, with an absolute difference of 0.13 Pa/s (34% relative discrepancy). Two factors should be considered when interpreting this result. First, the dataset was compiled from multiple test series conducted

on different material batches, introducing batch-to-batch variability that contributes to result dispersion. Second, the two devices differ considerably in vane height-to-diameter (H/D) ratio, with values of 0.10 and 2.5 for the MPSV-DIC and rheometer, respectively. The effect of such difference on structuration rate measurements of printable mortars is an open question: available experimental data only covers comparative yield stress measurements and report varying impact, which are moreover limited to narrower ranges of variation in the 1-3 range, and to different materials such as very flowable cement pastes [9,10] and soils [11]. In this context, the observed level of agreement is reasonable and suggests that both devices can yield comparable measurements when the cup size is sufficiently large. This is in line with similar findings of Barry et al. [12] who also documented good agreement between vane rheometer and PSV tests on printable mortars.

For the yield stress values, at similar ages, the rheometer small cup measurements values are notably increased compared to large cup ones. Also, the structuration rate obtained from small cup is 4.63 times higher than that obtained with the large cup. This substantial difference shows that the cup size may influence not only the yield stress, but also the apparent rate at which the material structure develops over time. This is likely an artifact, because the small cup failure pattern observed in Figure 2 is not cylindrical nor localized near the vane, suggesting that the Couette analogy framework on which the yield stress is computed is not valid here. This suggests boundary effects influencing the failure mechanism.

3.2 Influence zone of the pocket vane test: DIC observations of the strain field in the cup at yielding

To investigate the origin of these discrepancies, the strain field at the onset of yielding of a cement paste and a printable mortar was analyzed using MPSV-DIC.

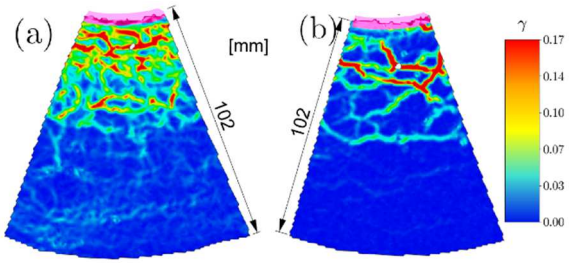


Fig. 4. Typical shear strain fields measured at the onset of yielding with cement paste (W/C = 0.3) at different ages, (a) 36 min and (b) 65 min.

Cement paste. Typical shear strain fields are illustrated in Figure 4 at the onset of yielding for cement pastes at different ages. The pink area highlights local strains occurring at the very edge of the vane, which correspond to the yielding event, typically within a 1 mm zone. These are difficult to measure with DIC because the region near the vane edge cannot be tracked as a material element. However, strains not only

localize at the vane boundary, but also extend into the specimen. Most significant strains (>0.1) are contained within an area which radial size is approximately equal to the vane diameter, which frames a main influence zone of the test. Strains tend to localize into diagonal and circumferential bands, which is relevant to the shear direction. These localizations become less diffuse with increasing material age, while the spatial extent of strains remains similar.

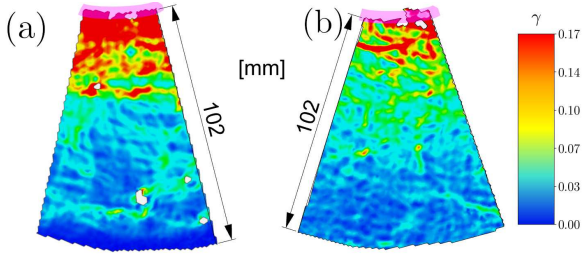


Fig. 5. Typical shear strain fields measured at the onset of yielding with the printable mortar, at different ages, (a) 31 min and (b) 51 min.

Printable mortar. As illustrated in Figure 5, the significant strains extend to the cup boundary. They are also more diffuse. This difference can be explained by the presence of granulates (sand), which might promote strain diffusion due to hard granular contacts. It is therefore plausible that having a boundary close to the vane, such as with the small cup rheometer, perturbs the strain field. The root causes may lie in the complex interplay among particle sizes, particle interactions (potentially affected by plasticizer), particle strength, and the cup wall surface properties.

4 Conclusion and Outlook

This study introduces the MPSV-DIC, a novel automated device for characterizing 3D printable mortars. Validation against rheometer tests confirms the reliability of the MPSV-DIC for yield stress and structuration rate measurements, provided the sample size is sufficiently large: a 126 mm sample diameter with a 48 mm vane diameter (39 mm gap) for the MPSV-DIC, or a 126 mm sample diameter with a 14 mm vane diameter (56 mm gap) for the rheometer. A small rheometer cup (29 mm diameter, 7.5 mm gap) overestimates yield stress and structuration, likely due to non-Couette failure from boundary interactions. MPSV-DIC analysis further reveals a broader influence zone for printable mortar than for cement paste, supporting the use of large cups and suggesting that the importance of this parameter may be related to mix design. Future work will assess strain measurement accuracy, explore varied cup sizes and printed specimens to further quantify the sample size effect, as well as for 2K (accelerated) printable mortars.

5 Acknowledgements

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